Teaching Statement

William Duan — Vancouver School of Economics, University of British Columbia

Every academic is a curator, responsible for a body of knowledge, for adding to it and for deciding what to put in front of people and how. Some of us are drawn to the dig, to being the first to lift something out of the ground and say what it is. Others are drawn to the David and the Last Supper, which they pore over as countless people have before them and then share with a public they hope to make as interested as they are. I enjoy both; the second is where I have done my best work. Anyone with a good memory can give a talk on the Last Supper from a brochure. The talk is different when it comes from someone who understands why the thing is beautiful, and that difference is the whole case for having the people who understand a subject be the ones who teach it.

I have taught economics at every level UBC offers, from first-year principles to the PhD core, over seven years as a teaching assistant and this past summer as instructor of the department's graduate Math Camp, whose PhD sessions I designed. My research is in mechanism design and auction theory. Teaching is the part of academic work I enjoy most, and I want it to stay central to whatever I do next.

Teaching interests

I would be glad to teach any part of the core undergraduate sequence, macroeconomics and econometrics included. The core is where most students meet economics for the first and often the last time, which makes it the most consequential teaching a department does. My strongest preference and deepest preparation are in microeconomics, where I can teach any undergraduate course from principles through game theory, information economics, and auctions and market design, and at the graduate level mathematics for economists and the microeconomic theory core.

Given the chance, the course I would most like to build is an upper-year course on how economics separated from philosophy and the other social sciences and acquired its characteristic way of reasoning. Each unit would take a question that preoccupied the social sciences of its period, set out what philosophers and other social scientists made of it, and ask what economists did differently and why. That structure works until roughly the 1960s, when economists stop defining themselves against philosophers and sociologists and start defining themselves against earlier economists. The course would end there, since that is the moment a discipline is fully born.

Teaching philosophy

Underneath all of this is an assumption about the division of labour between student and teacher. At a university, students lead their own learning. The teacher's job is narrower than it looks and has three parts: tell students clearly what to learn and to what level, help them over the rough patches, and give them a reason to want to. What follows is how I do each.

Of the three, the first is the one I consider the most important. Set the expectation clearly, then make sure students have everything they need to meet it. In a course built on models, the expectation that matters most is which results students should be able to derive and which they should be able to use. That line has to be drawn early and kept, with nothing bleeding across it. A result on the derived side needs the complete derivation, carefully done. A result on the given side needs to be stated plainly as an assumption, and nothing more. Bleed in either direction does the damage. A half-finished derivation of something meant to be taken on trust, or no label at all, leaves students unable to tell whether the gap is

in the argument or in themselves, and invites them to spend time deriving something they may not be equipped to derive. An object they are meant to build on, handed over as a picture, leaves nothing built on it secure. Where the line sits depends on who is in the room. A first-year undergraduate and a first-year PhD student are looking at the same object; they are equipped to see different parts of it.

In an introductory course the derived side is small and has to be something students can build completely from what they already have. For me that is the economic way of thinking itself, a discipline they can practise on situations they know: take one, remove detail until what remains is simple enough to reason about, and check that it still captures the tension you started with. Every step of that can be done without mathematics, which is why the mathematics stays in the background. The idea surprises many students, who arrive believing that rigour means including everything, and it is the idea worth showing them.

In upper-year and graduate courses the derived side is most of the course. Students with the mathematical background understand a result best by deriving it, and I have found that they are stuck most often not on the hard theorems but on objects handed to them without construction. The lottery simplex is the usual example. It is introduced as a picture and used for the rest of the term, and students who cannot say where it came from struggle with everything built on it. In an ECON 600 tutorial I derived it from scratch instead, starting from the fact that any lottery over three outcomes is a point on a triangular surface in three-dimensional space and showing that the simplex is that surface, rescaled and renormalized. It is not a deep argument. One student wrote afterwards that it made sense of something they had not fully understood for years.

Teaching practice

My sections are built around problems rather than lecture. One exercise I rely on in the PhD core grew out of an older one. For some time I have presented proofs that are correct in their approach but omit a few necessary and subtle steps, and asked students to find where the argument is incomplete. This year I have started generating flawed proofs with weaker AI models and asking students to locate the mistake. Reading a plausible argument and finding where it fails is among the most valuable skills a graduate student can take from a theory course, and lectures do not train it.

Much of the teaching happens outside the room, and it depends on the relationship students have with me and with each other. In a class of twenty I work at making students comfortable with me, through a casual manner and an interest in their lives beyond the course, so that they will bring me even their small questions. In a large class the useful connection is among the students. In ECON 101 I put the whole section into one group chat and routed every question through it. Most were answered by classmates before I saw them, and my job became confirming the answer or adding a small correction. Students who would never have emailed a teaching assistant asked freely there, and the group made me more approachable than office hours alone could.

Office hours are the other half of that. I hold one and a half to two hours a week, more before exams, and they routinely run long. When a student brings a problem, I try to reconstruct how they were thinking about it before I say anything, because the useful answer is rarely the correct solution alone but the correct solution attached to the place where their reasoning went wrong. With PhD students, office hours often drift into their own research questions, which I am always glad to discuss. My evaluations have changed how I run tutorials. For several years they praised the intuition and the range of angles while noting, fairly, that I improvise too much; an alternative route I find elegant is not what a student

taking their first steps needs. I now come to every tutorial with a clean, linear solution written out in advance, present that first, and save the alternatives for the end.

Students in any UBC section differ widely in preparation, in confidence, and in whether English is their first language, and the habits I have settled into are mostly answers to that range. I explain the same idea from several angles rather than one angle more slowly, so that a student who missed the first route has a second. In office hours everyone asks once before anyone asks twice, so that the quiet students are not crowded out by the confident ones. I try never to assume a student knows more than they should at that point in the course, and never to make them feel small for knowing less. I would carry the same habits into large first-year classes, where the range is wider and the quiet students easier to lose.

Evidence of teaching effectiveness

I have been the teaching assistant for UBC's PhD microeconomic theory sequence (ECON 600 and 601) every year since 2021. Across all ten sections, 100% of responding students rated my overall contribution favourably, with interpolated medians between 4.8 and 5.0 out of 5. I received the Vancouver School of Economics (VSE) Agustín Beltrán Award for Excellence in Teaching, the department's award for teaching assistants, in 2024, 2025 and 2026. Before that I taught discussion sections for ECON 101, 303, 355 and 514 (principles, intermediate microeconomics, international trade, and information and incentives). In Summer 2026 I taught the MA and PhD sessions of the VSE Math Camp. I designed the PhD sessions with Jesse Perla and wrote their slides and practice problems; the MA sessions ran on recorded faculty lectures, with my part being the problem sessions and questions. Complete evaluations for every term I have taught, with a summary table, and the full Math Camp materials are linked from my website.

Curriculum work and plans

Beyond my own sections, I co-designed problem sets for ECON 601 and designed the PhD sessions of the Math Camp, deciding with Jesse Perla what to keep from earlier material and what to build new; the department intends to keep the result as the basis for future instructors. One ambition runs through what I would like to do next. Most students in an economics classroom will not go on in economics, and the most valuable thing the subject can give them is its way of thinking, which is best appreciated against other ways of thinking. In introductory courses I would like to show, alongside the economic analysis of a problem, how a sociologist, a historian or a philosopher would approach it, so that what is distinctive about ours is visible by contrast. The origins course described above is the same idea carried through a full term.